communication, control architecture, and stability management concepts that would later evolve into the sophisticated autonomous systems used today. In essence, these early systems transformed the theoretical idea of pilotless aircraft into a tangible reality, providing the first stepping stone toward the modern era of intelligent, data-driven aerial robotics.

Thus, the first generation of drones symbolizes not only the birth of unmanned flight but also the beginning of a technological revolution that continues to shape global defense, commercial, and research landscapes. These early experiments laid the groundwork for subsequent generations of drones that would incorporate automation, artificial intelligence, and advanced navigation, transforming simple manually controlled aircraft into the highly capable autonomous systems that define the field today.

2.1.1 Definition

First-generation drones refer to the earliest form of unmanned aerial vehicles (UAVs) developed primarily during the early to mid-20th century, charac-terized by manual radio-controlled operation, limited range, and rudimen-tary flight stability. These drones were designed mainly for military target practice, surveillance, and experimental purposes, serving as prototypes for future unmanned flight systems. They lacked onboard intelligence, sensors, and automated navigation, operating entirely under human control from the ground via analog radio transmitters.

2.1.2 Basic characteristics of first-generation drones

The first generation of drones represents the earliest stage in the evolution of unmanned aerial systems (UAS), characterized by simple mechanical structures and entirely manual control mechanisms. These drones were primarily used during the early to mid-20th century for military training, target simulation, and limited reconnaissance missions. Although their